

CHE 301 Hw 6

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Abstract

This assignment explores advanced thermodynamic principles through four distinct problems focused on real-world applications. Problem A involves deriving a thermodynamic identity that connects compressibility and residual properties. Problem B requires a detailed procedure to calculate the fugacity of dry ice under specified conditions using the Peng-Robinson equation of state. Problem C examines the production of dry ice via a throttling process, calculating the mass fraction formed during expansion. The problems collectively reinforce core concepts such as fugacity, phase behavior, and thermodynamic modeling, while encouraging the use of computational tools and critical assumptions.

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1 Problem A

This problem was solved utilizing ChatGPT. We want to prove:

$$\int_0^P (1 - \beta T) V dP = RT(Z - 1) - \int_0^P RT^2 \left. \frac{\partial Z}{\partial T} \right|_{\rho} \frac{d\rho}{\rho}$$

where

$$Z = \frac{PV}{RT}, \quad \beta = \frac{1}{V} \left(\frac{\partial V}{\partial T} \right)_P$$

and ρ is the molar density.

Step 1: Express $(1 - \beta T)V$ in terms of Z

We have $V = \frac{ZRT}{P}$. Then:

$$\begin{aligned} \left(\frac{\partial V}{\partial T} \right)_P &= \frac{R}{P} \left[Z + T \left(\frac{\partial Z}{\partial T} \right)_P \right] \\ \beta &= \frac{1}{V} \cdot \frac{R}{P} \left[Z + T \left(\frac{\partial Z}{\partial T} \right)_P \right] = \frac{V}{ZT} \cdot \frac{1}{V} \left[Z + T \left(\frac{\partial Z}{\partial T} \right)_P \right] = \frac{1}{ZT} \left[Z + T \left(\frac{\partial Z}{\partial T} \right)_P \right] \\ \beta &= \frac{1}{T} + \frac{1}{Z} \left(\frac{\partial Z}{\partial T} \right)_P \end{aligned}$$

Therefore:

$$1 - \beta T = 1 - \left[1 + \frac{T}{Z} \left(\frac{\partial Z}{\partial T} \right)_P \right] = -\frac{T}{Z} \left(\frac{\partial Z}{\partial T} \right)_P$$

So the LHS becomes:

$$\text{LHS} = \int_0^P \left[-\frac{T}{Z} \left(\frac{\partial Z}{\partial T} \right)_P \right] V dP = -RT^2 \int_0^P \frac{1}{P} \left(\frac{\partial Z}{\partial T} \right)_P dP$$

Step 2: Change variables from P to ρ

We have $P = Z(T, \rho) \rho RT$. At constant T :

$$\begin{aligned} dP &= RT \left[Z + \rho \left(\frac{\partial Z}{\partial \rho} \right)_T \right] d\rho \\ \frac{dP}{P} &= \frac{RT [Z + \rho Z_{\rho}] d\rho}{Z \rho RT} = \frac{Z + \rho Z_{\rho}}{Z \rho} d\rho \end{aligned}$$

We need $\left(\frac{\partial Z}{\partial T} \right)_P$ in terms of $\left. \frac{\partial Z}{\partial T} \right|_{\rho}$:

$$\left(\frac{\partial Z}{\partial T} \right)_P = \frac{Z}{Z + \rho Z_{\rho}} \left[Z_{T\rho} - \frac{\rho Z_{\rho}}{T} \right]$$

where $Z_{T\rho} = \left. \frac{\partial Z}{\partial T} \right|_{\rho}$.

Step 3: Combine the factors

The integrand becomes:

$$\frac{1}{P} \left(\frac{\partial Z}{\partial T} \right)_P dP = \frac{Z + \rho Z_\rho}{Z\rho} \cdot \frac{Z}{Z + \rho Z_\rho} \left[Z_{T\rho} - \frac{\rho Z_\rho}{T} \right] d\rho = \frac{1}{\rho} \left[Z_{T\rho} - \frac{\rho Z_\rho}{T} \right] d\rho$$

Thus:

$$\text{LHS} = -RT^2 \int_0^\rho \frac{1}{\rho} \left[Z_{T\rho} - \frac{\rho Z_\rho}{T} \right] d\rho$$

Step 4: Separate the integral

$$\text{LHS} = -RT^2 \int_0^\rho \frac{Z_{T\rho}}{\rho} d\rho + RT \int_0^\rho Z_\rho d\rho$$

The second term:

$$RT \int_0^\rho Z_\rho d\rho = RT[Z(\rho) - Z(0)] = RT(Z - 1)$$

since $Z(0) = 1$ for ideal gas at zero density.

Therefore:

$$\text{LHS} = RT(Z - 1) - RT^2 \int_0^\rho \frac{\partial Z}{\partial T} \Big|_\rho \frac{d\rho}{\rho}$$

This matches the RHS, completing the proof.

2 Problem B

In this problem, we are tasked to write a procedure to calculate the fugacity of dry ice at pressure 100 atm and temperature -120°C . We are assuming that the dry ice is 100% CO_2 , and its vapor properties follow the Peng-Robinson Equation of State.

To start, we use the Clausius-Clapeyron thermodynamic relationship:

$$\ln \left(\frac{P_2}{P_1} \right) = \frac{\Delta H_{\text{sub}}}{R} \left(\frac{1}{T_1} - \frac{1}{T_2} \right)$$

This equation is derived from the following:

$$\frac{dP}{dT} = \frac{\Delta H_{\text{sub}} P}{RT^2}$$

$$\frac{dP}{P} = \frac{\Delta H_{\text{sub}}}{RT^2} dT$$

$$\int_{P_1}^{P_2} \frac{dP}{P} = \int_{T_1}^{T_2} \frac{\Delta H_{\text{sub}}}{RT^2} dT$$

$$\ln\left(\frac{P_2}{P_1}\right) = \frac{\Delta H_{\text{sub}}}{R} \left(\frac{1}{T_1} - \frac{1}{T_2}\right)$$

The bounds P_1 and T_1 represent the normal sublimation pressure and temperature of CO_2 , which are $P_1 = 1$ atm and $T_1 = -78.5$ °C, as can be seen in Figure 1. P_2 and T_2 represent the pressure and temperature at -120 °C on the saturation curve of the Pressure-Temperature diagram. Along this curve, the fugacity of the vapor state of carbon dioxide is equal to the fugacity of the solid state, which means we can use vapor properties to determine the fugacity of the solid. We will solve for P_2 using the Clasius-Clapeyron equation above.

$$P_2 = P_1 \exp\left[\frac{\Delta H_{\text{sub}}}{R} \left(\frac{1}{T_1} - \frac{1}{T_2}\right)\right]$$

H_{sub} is the enthalpy of sublimation, which for carbon dioxide is 26000 kJ/k-mol. The exponent is unitless when the R gas constant is in kJ/kmol K, and temperature in K, so P_2 will always be in the units of whatever P_1 is. In our case, we used bar:

$$P_2 = 1.013 \exp\left[\frac{26000}{8.314} \left(\frac{1}{194.65} - \frac{1}{153.15}\right)\right] = 0.0130\text{bar}$$

So, the sublimation pressure at -120 °C for carbon dioxide is 0.0130 bar. We will use this pressure and temperature to calculate the fugacity of the vapor because

$$f_v(T = -120^\circ\text{C}, P_{\text{sub}} = P_2) = f_s(T = -120^\circ\text{C}, P_{\text{sub}} = P_2)$$

To calculate the fugacity f_v , we must first determine the fugacity coefficient ϕ at P_2 and T_2 . This is done using the CHE Calculator function `=PhiC`, with the following inputs: CO_2 as the component, a temperature of -210 °C, P_2 as the pressure, and the compressibility factor Z of the vapor.

The Z value was obtained using the function `=ZComp`, set to the Peng–Robinson equation of state. The inputs for `=ZComp` were CO_2 as the component, a temperature of -120 °C, P_2 as the pressure, and an initial guess of $Z = 1$, since the substance is in the vapor phase.

The resulting values for Z and ϕ are as follows:

$$Z = 0.9995638 \quad \text{and} \quad \phi = 0.9995640$$

Fugacity is expressed in the following relationship:

$$f_v = \phi P_2 = (0.9995640)(0.0130\text{bar}) = 0.013029\text{bar}$$

So,

$$f_v(T = -120^\circ\text{C}, P_{\text{sub}} = P_2) = f_s(T = -120^\circ\text{C}, P_{\text{sub}} = P_2) = 0.013029\text{bar}$$

A correction factor must be applied to find the fugacity at -120 °C and 100 atm. This is given by the following:

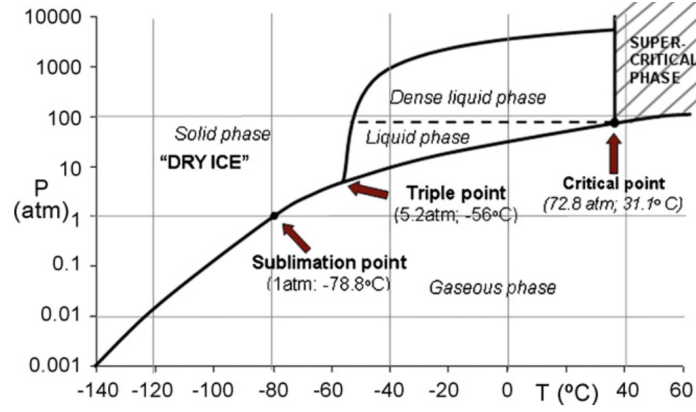


Figure 1: P-T diagram for CO₂
 Reproduced from Mazzoldi, A., & Hill, T. A. (2008),
 DOI:10.1016/S1750-5836(07)00118-1

$$\begin{aligned}
 f_s(-120^\circ\text{C}, 100 \text{ atm}) &= f_s(-120^\circ\text{C}, P_2) \exp \left(\int_{P_2}^{100 \text{ atm}} \frac{\bar{v}_s}{RT} dP \right) \\
 &= f_s(-120^\circ\text{C}, P_2) \exp \left[\frac{\bar{v}_s}{RT} (100 \text{ atm} - P_2) \right]
 \end{aligned}$$

where $\bar{v}_s$ is the molar volume = 0.0282 m³/kmol, according to ChatGPT. Plugging our values in, we get the following result:

$$f_s = 0.013059 \text{ bar} = 0.012888 \text{ atm}$$

3 Problem C

In this problem, we need to determine the mass fraction of dry ice (CO₂(s)) formed through an expansion in a throttling valve.

Given Conditions

- Inlet pressure: $P_1 = 60 \text{ atm} = 60.795 \text{ bar}$
- Inlet temperature: $T_1 = -35^\circ\text{C}$
- Outlet pressure: $P_2 = 1.01325 \text{ bar}$
- Outlet temperature: $T_2 = -78.5^\circ\text{C}$ (sublimation temperature from Part 2)

Procedure

Before beginning calculations, the equation of state (EOS) must be changed to the Peng–Robinson EOS.

Step 1: Compressibility Factors

The compressibility factor Z is computed using the `ZComp` function, with an initial guess of $Z = 1$.

$$Z_{\text{in}} = 0.116953 \quad \text{and} \quad Z_{\text{out}} = 0.981401$$

Step 2: Solid Fraction Assumption

Since this process involves solid formation rather than vapor–liquid equilibrium, the solid fraction is used instead of a vapor fraction. For the inlet stream, the solid fraction is:

$$x_{\text{solid,in}} = 0$$

because the entire stream is vapor. For the outlet stream, the solid fraction $x_{\text{solid,out}}$ will be greater than zero. An initial guess of $x_{\text{solid,out}} = 0.5$ is used.

Step 3: Enthalpy Balance

For a throttling valve, the energy balance gives:

$$H_{\text{in}} = H_{\text{out}}$$

The inlet enthalpy is calculated using the Peng–Robinson EOS with the `Enthalpy` function:

$$H_{\text{in}} = H_{\text{vapor,in}} = -410599 \text{ kJ/kmol}$$

For the outlet stream, both vapor and solid phases are considered. The enthalpy of the solid phase can be determined using the thermodynamic identity:

$$H_{\text{solid}} = H_{\text{vapor}} - H_{\text{sublimation}}$$

From Part 2, the enthalpy of sublimation is given as:

$$H_{\text{sublimation}} = 26000 \text{ kJ/kmol}$$

$$H_{\text{solid}} = -397440 - 26000 = -423440 \text{ kJ/kmol}$$

The vapor enthalpy at the outlet, $H_{\text{vapor,out}}$, is computed using the same method as for the inlet.

Applying the lever rule for a solid–vapor mixture:

$$H_{\text{out}} = x_{\text{solid,out}} H_{\text{solid}} + (1 - x_{\text{solid,out}}) H_{\text{vapor,out}}$$

	S1	S2		Hsub kJ/kmol	26000
P bar	60.795	1.01325			
T C	-35	-78.5	<-- sublimation point		
Z	0.116953	0.981401			
solid fraction	0	0.506125	<--- percentage that solidified		
H vapor	-410599	-397440			
H solid	0	-423440			
Enthalpy	-410599	-410599			
Enthalpy Balance	0				

Figure 2: Excel Sheet for Part C

Step 4: Iterative Solution

To satisfy the energy balance, the difference between inlet and outlet enthalpies is minimized:

$$H_{\text{in}} - H_{\text{out}} = 0$$

Using the GoalSeek function, this equation is solved by varying $x_{\text{solid,out}}$.

Result

Through this iterative process, the mass fraction of solidified CO₂ (dry ice) is determined to be:

$$x_{\text{solid,out}} = 0.506125$$

Approximately 50.6% of the CO₂ stream solidifies upon expansion through the throttling valve under the given conditions.

4 Conclusion

This assignment deepens understanding of thermodynamic systems by applying theoretical models to practical scenarios involving CO and organic mixtures. The derivation in Problem A strengthens analytical skills in manipulating thermodynamic identities. Problems B and C highlight the utility of equations of state in predicting phase transitions and fugacity, essential for chemical process design. Overall, the assignment integrates conceptual rigor with engineering relevance, preparing students for complex thermodynamic analysis in both academic and industrial contexts.